Name:

BUID:

Workshop 2: Matrices of Linear Transformations

1. Suppose T is a linear transformation with the following input-output behavior: $T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = 9 \qquad T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = 2$

   (a) Determine the domain and codomain of T. (b) Determine the value of $T\left(\begin{bmatrix}2\\-3\end{bmatrix}\right)$.

2. Suppose T is a linear transformation with the following input-output behavior: $T\left(\begin{bmatrix}1\\2\end{bmatrix}\right) = \begin{bmatrix}3\\4\end{bmatrix} \qquad T\left(\begin{bmatrix}3\\4\end{bmatrix}\right) = \begin{bmatrix}5\\6\end{bmatrix}$

   (a) Determine the domain and codomain of T.

   (b) Determine the following values: $T\left(\begin{bmatrix}4\\6\end{bmatrix}\right) \quad T\left(\begin{bmatrix}2\\3\end{bmatrix}\right) \quad T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) \quad T\left(\begin{bmatrix}0\\1\end{bmatrix}\right)$

   (c) Determine the matrix that implements T. (d) Is the $\mathrm{ran}(T) = \mathrm{cod}(T)$? Explain your answer.

3. Determine the matrix that implements rotation by 90 degrees clockwise about the origin in $\mathbb{R}^2$.

4. Determine the matrix that implements the following transformation: $\begin{bmatrix}x_1\\x_2\end{bmatrix} \mapsto \begin{bmatrix}x_1+x_2\\x_1\\x_2-2x_1\end{bmatrix}$

5. Determine the matrix that implement rotation about the z-axis in $\mathbb{R}^3$ counterclockwise 180 degrees, followed by reflection across the xy-plane.

6. Let A be the matrix $\begin{bmatrix}1 & -1\\1 & 2\end{bmatrix}$. Draw the the image of the unit square under the transformation $\mathbf{x} \mapsto A\mathbf{x}$.

7. Determine a matrix B, such that $A \neq B$ and the image of the unit square under the tranformation $\mathbf{x} \mapsto B\mathbf{x}$ is the same as the image under $\mathbf{x} \mapsto A\mathbf{x}$ (i.e., if you replace A with B in the previous problem, you would get the same drawing).

Write your solutions below and on the back on this page.

1a. $\mathrm{dom}(T) = \mathbb{R}^2$ $\quad \mathrm{cod}(T) = \mathbb{R}$

1b. $T\left(\begin{bmatrix}2\\-3\end{bmatrix}\right) = T\left(2\begin{bmatrix}1\\0\end{bmatrix} - 3\begin{bmatrix}0\\1\end{bmatrix}\right) =$

$2T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) - 3T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) =$

$2(9) - 3(2) = 18 - 6 = 12$

2a. $\mathrm{dom}(T) = \mathbb{R}^2$ $\quad \mathrm{cod}(T) = \mathbb{R}^2$

b. $T\left(\begin{bmatrix}4\\6\end{bmatrix}\right) = T\left(\begin{bmatrix}1\\2\end{bmatrix} + \begin{bmatrix}3\\4\end{bmatrix}\right) =$

$T\left(\begin{bmatrix}1\\2\end{bmatrix}\right) + T\left(\begin{bmatrix}3\\4\end{bmatrix}\right) = \begin{bmatrix}3\\4\end{bmatrix} + \begin{bmatrix}5\\6\end{bmatrix} = \begin{bmatrix}8\\10\end{bmatrix}$

$T\left(\begin{bmatrix}2\\3\end{bmatrix}\right) = T\left(\frac{1}{2}\begin{bmatrix}4\\6\end{bmatrix}\right) = \frac{1}{2}T\left(\begin{bmatrix}2\\3\end{bmatrix}\right)$

$= \frac{1}{2}\begin{bmatrix}8\\10\end{bmatrix} = \begin{bmatrix}4\\5\end{bmatrix}$

$T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) = T\left(2\begin{bmatrix}1\\2\end{bmatrix} - \begin{bmatrix}2\\3\end{bmatrix}\right)$

$= 2T\left(\begin{bmatrix}1\\2\end{bmatrix}\right) - T\left(\begin{bmatrix}2\\3\end{bmatrix}\right) =$

$2\begin{bmatrix}3\\4\end{bmatrix} - \begin{bmatrix}4\\5\end{bmatrix} = \begin{bmatrix}2\\3\end{bmatrix}$

$T\left(\begin{bmatrix}1\\0\end{bmatrix}\right) = T\left(\begin{bmatrix}1\\2\end{bmatrix} - 2\begin{bmatrix}0\\1\end{bmatrix}\right) =$

$T\left(\begin{bmatrix}1\\2\end{bmatrix}\right) - 2T\left(\begin{bmatrix}0\\1\end{bmatrix}\right) =$

$\begin{bmatrix}3\\4\end{bmatrix} - 2\begin{bmatrix}2\\3\end{bmatrix} = \begin{bmatrix}-1\\-2\end{bmatrix}$

(c) $\begin{bmatrix}-1 & 2\\-2 & 3\end{bmatrix} \sim \begin{bmatrix}-1 & 2\\0 & -1\end{bmatrix}$

(d) Yes One pivot per row.

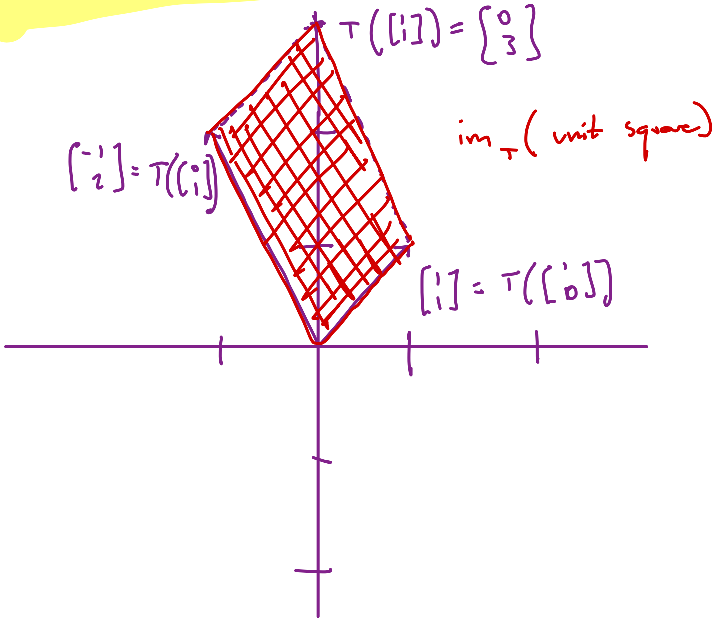
3. $\begin{bmatrix}0 & 1\\-1 & 0\end{bmatrix}$

Continue your work on the here.

4. $\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ -2 & 1 \end{bmatrix}$

5. $\begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$

6.



7. $\begin{bmatrix} -1 & 1 \\ 2 & 1 \end{bmatrix}$